7 Conclusion

Under rightward traction, each clamped object extends and contracts sideways. For the square, the mean right-edge horizontal displacement approaches about 0.494 mm, with decreasing changes under refinement. Geometry and supports change this response in a clear way: the rectangle of twice the length extends twice as far, the central opening increases the extension by 13% because the load must pass around it, and clamping only half of the left edge increases it by 11 to 25%. Moving the load along the edge changes the bending, not the mean extension.

For the supported symmetric positive-definite system, conjugate gradients is a practical starting choice among the tested iterative methods. It reaches the required residual with much less work than Gauss–Seidel on the square and on every other tested object. Diagonal preconditioning reduces the CG iteration count slightly, but does not improve the measured time, so it is not an automatic improvement. Ordinary Jacobi diverges, while damped Jacobi and steepest descent fail to reach the target within the iteration budget, and Gauss–Seidel becomes impractical as the condition number grows. These findings support checking convergence and total cost rather than selecting a method by simplicity or iteration count alone. The untimed sparse-LU solution remains a useful same-mesh reference, not a measured performance competitor.

The validation checks answer different questions. Prescribed displacements, force and moment balance, and agreement with the direct reference support the discrete solution. Refinement tests its dependence on the mesh; the separate exact affine test checks a different, roller-supported case. Neither establishes the continuum error of the clamped square. In addition, conditioning worsens as the mesh is refined even while the measured displacements settle. The load-perturbation tests show that sensitivity also depends on how the load changes: the condition number describes possible amplification, not the response to every perturbation.

These conclusions are limited to the small-deformation, plane-stress model and the three tested objects. The condition number grows with mesh refinement, with the distance between support and load, and when the support is shortened, so finer meshes or larger objects call for a preconditioned Krylov method. Near sharp corners, where stresses concentrate, a locally refined mesh would be needed before the computed stresses could be trusted.